Design and Evaluation of a Hierarchical Memory Architecture for Improving Long-Term Coherence in Local Large Language Model Dialogue Systems

Context

Large Language Models (LLMs) have demonstrated strong capabilities in generating coherent and contextually relevant natural language responses. These models are increasingly used in conversational applications including virtual assistants, tutoring systems, and software development support tools (Vaswani et al., 2017; Touvron et al., 2023). The availability of open-source models allows such systems to be deployed locally providing greater control and customisation to the LLM so that a user can change it to try and improve it compared to any cloud-based alternatives (Touvron et al., 2023).

Despite their capabilities, LLMs have a fundamental limitation due to their reliance on fixed-length context windows. Information that falls outside this window becomes inaccessible to the model, preventing it from retaining knowledge across extended interactions or multiple sessions. This limitation reduces the ability of dialogue systems to maintain long-term coherence and retain user-specific information. As a result, users must repeatedly provide the same information, reducing usability and efficiency. Bubeck et al. (2023) highlights that while modern LLMs do not possess reliable mechanisms for long-term memory retention or consistent knowledge persistence across interactions.

Current approaches such as sliding context windows and retrieval-augmented generation, which integrates external memory by retrieving relevant information from a vector database and inserting it into the model's context before response generation (Lewis et al., 2020). This provides partial solutions but typically treat memory as a flat structure without differentiation between short-term and long-term information, resulting in retrieval of irrelevant or redundant information. Therefore, the idea that LLMs could become more like humans when it comes to memory management and humans' long-term retention (Baddeley, 1992) to help improve the LLMs is evident, and would be useful as a concept for building a framework for LLMs.

My project aims to investigate whether introducing a hierarchical memory architecture into a locally deployed Llama 3.1 8B dialogue system can improve long-term dialogue coherence and information retention. This system will store and retrieve structured memory across interactions and will be experimentally compared against a baseline system without persistent memory. This comparison will allow me to evaluate whether hierarchical memory provides measurable improvements in dialogue consistency, detecting errors, and lowering down hallucination. The results will contribute to understanding how structured memory architectures can enhance the performance and reliability of locally deployed LLM dialogue systems.

Key Background Sources

Source	Description	Relationship to Project
Vaswani et al. (2017) – Attention Is All You Need	Introduces the transformer architecture, which uses self-attention mechanisms to model relationships between tokens in a sequence. This architecture forms the foundation of modern Large Language Models (LLMs).	Provides foundational understanding of how LLMs process language and generate responses.
LLaMA: Open and Efficient Foundation Language Models	Describes the design, training, and performance of LLaMA models, demonstrating that open-source LLMs can achieve performance comparable to proprietary models while remaining efficient enough for local deployment.	Provides technical background for the model I'll use in this project without relying on cloud infrastructure.
Lewis et al. (2020) – Retrieval-Augmented Generation	Introduces retrieval-augmented generation, which enhances language models by retrieving relevant information from external vector databases and incorporating it into the context during response generation.	Provides a foundation for implementing episodic memory retrieval in this project using vector similarity search. Demonstrates how external memory can improve factual accuracy and information access.
Johnson et al. (2017) – Billion-Scale Similarity Search with GPUs	Introduces FAISS, a highly efficient vector similarity search library capable of storing and retrieving high-dimensional embeddings at large scale with low latency.	For the implementation of episodic memory storage and retrieval in the proposed hierarchical memory system and provides the technical mechanism efficiently retrieve relevant past interactions.
Baddeley (1992) – Working Memory	Presents a cognitive model of human memory consisting of multiple components, including working memory, episodic memory, and semantic memory, each serving distinct functional roles.	Provides a theoretical foundation for designing a hierarchical memory architecture inspired by human cognitive memory structures.
Bubeck et al. (2023) – Sparks of Artificial General Intelligence	Evaluates advanced capabilities of modern LLMs and highlights both their strengths and limitations, including inconsistency, hallucination, and lack of persistent long-term memory.	Provides evidence that modern LLMs lack reliable long-term memory and consistent retention of information across interactions.

Aim

My aim is to implement and evaluate a hierarchical memory architecture to improve long-term coherence and retention in a locally deployed Large Language Model dialogue system.

Objectives

1. Implement a baseline dialogue system using a locally deployed LLaMA 3.1 8B model with standard context-window-based memory, capable of maintaining conversations across at least 10 dialogue turns without persistent external memory.
2. Design and implement a hierarchical memory architecture consisting of:
    - Working memory for recent conversation context
    - Episodic memory for storing past interactions using vector similarity search
    - Semantic memory for storing persistent user information
3. Develop a memory controller mechanism to determine which information should be stored and consolidated.
4. Integrate the hierarchical memory system into the dialogue pipeline.
5. Design and conduct controlled experimental dialogue scenarios consisting of at least 5 multi-session conversations, where specific information is introduced and later tested to evaluate memory retention and retrieval performance.
6. Evaluate performance using measurable criteria including:
    - Retention accuracy (Percenttage of correctly recalled information)
    - Dialogue coherence (measured by consistency of responses across sessions)
    - Consistency across interactions (Percentage of relevant memories successfully retrieved)
    - Frequency of forgotten information (number of failed recall instances per conversation)
    - Error detection
    - Frequency of hallucination
7. Analyse the results to determine whether hierarchical memory improves dialogue performance and identify the limitations of the system.

Planning

TASK	START	END
Literature Review		
Reasearch Paper on Working Memory	9/2/26	12/2/26
Reasearch Paper on Open and efficient foundation language models	12/2/26	14/2/26
Reasearch Paper on Different LLM to use	14/2/26	18/2/26
Baseline System		
Implement LLM	22/2/26	26/2/26
Ensure baseline functional system	24/2/26	1/3/26
Memory System Development	1/3/26	15/3/26
System Testing	3/3/26	3/19/26
System Evaluation	19/3/26	27/3/26
Dissataion Writeup	1/3/26	6/4/26

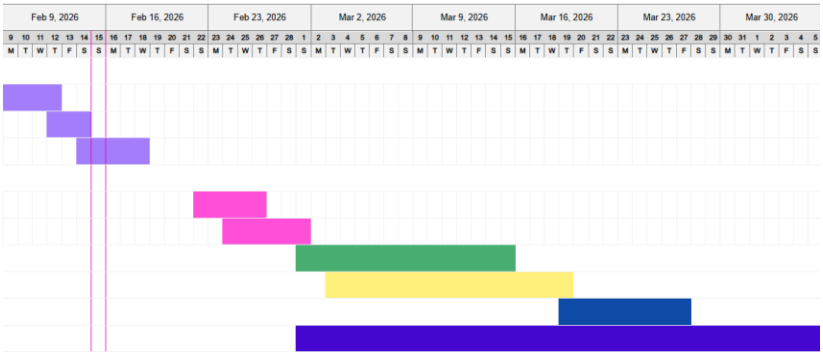

Figure 1: Project Gantt Chart

As seen in Figure 1, I started by reading literature reviews to build a clear understanding of LLMs and memory systems before the development starts on my system. The baseline system is built first so there is something to compare against later.

I will build a baseline dialogue system which will be implemented first to provide a reference point for comparison. This enables objective evaluation of the system and whether the hierarchical memory architecture has improved performance.

I will then develop a new memory architecture and integrated gradually to reduce complexity and make testing easier.

I will carry out experiments after both systems are complete so their performance can be compared fairly, and from this carry out an evaluation on both systems. I will also be writing my dissertation alongside the project to ensure progress is properly recorded and to avoid leaving everything until the end.

Risks

Hardware limitations may affect model performance or responsiveness. This will be mitigated using model optimisation techniques such as quantisation.

Memory retrieval may introduce irrelevant or incorrect information, potentially reducing system performance. The retrieval methods will be tested and refined to improve relevance.

Error detection and hallucination reduction may be difficult for me to measure reliably. I will mitigate this by designing controlled dialogue scenarios with known facts so incorrect responses can be clearly identified and compared between systems.

Integration complexity may increase my development time. This will be mitigated by implementing and testing each component independently before integration.

Experimental evaluation may give me inconclusive results. Multiple test scenarios will be used to give me reliable and meaningful evaluation.

References

Baddeley, A.D., 1992. Working memory. *Science*, 255(5044), pp.556–559.

Bubeck, S. et al., 2023. Sparks of artificial general intelligence: Early experiments with GPT-4. *arXiv preprint arXiv:2303.12712.*

Johnson, J., Douze, M. and Jégou, H., 2017. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3), pp.535–547.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T. and Riedel, S., 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33.

Touvron, H. et al., 2023. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971.*

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. *Advances in Neural Information Processing Systems*, 30.